Experiment 11a

2/10/24 Simulate static routing configuration using cisco packet tracer.

Aim: To Simulate Static routing configuration.

the process of adding static routes to the routing table is known as static Routing.

Router 0	10.0.0.0/8, 20.0.0.0/8, 40.0.0.0/8	30.0.0/8, 50.0.0.0/8
Router 1	20.0.0.0/8 30.0.0.0/8 50.0.0.0/8	10.0.0.0/8, 40.0.0.0/8
Router 2	40.0.0.0/8 50.0.0.0/8	10.0.0.0/8, 20.0.0.0/8, 30.0.0.0/8

Router 0 Requirements

* Create 2 routes for network 30.0.0.0/8, configure the first route as the main route, second route as a backup route
* create 2 routes for the host 30.0.0.100/8 configure the first route as main route second route as a backup one.
* Using the router adds only main route to

routing table

ROUTER 0 configuration

ROUTER > enable

ROUTER # configure

terminal

Enter configuration commands, one per line End with CNTL/Z.

ROUTER (config) # ip route 30.0.0.0 255.0.0.0 20.0.0.2

ROUTER (config) # exit

ROUTER # show ip route static.

30.0.0.0/8 is variably subnetted, 2 subnets, 2

mask S 30.0.0.0/8 [10/10] via 20.0.0.2

S 30.0.0.100/32 [10/10] via

Similarly we can configure the Router 1, 2 requirements and we can configure the Router 1, 2.

Verifying Static Routing:

On Router 0, we configured 2 routes for network 30.0.0.0/8. These Routes via Router 1 and No Route 2. First Route as main route, Second route as backup route.

→ By sending ping request from PC 4 network 30.0.0.0/8, and tracing the path we can use tracert command

→ By clearing the routing table entries on Router 0.

to delete Static Route

→ use 'show ip route static'

→ note down routes to be deleted

→ Use the no 'ip route' command to delete the route.

Result:

Thus Static Routing has been implemented.

27/8/26

Ex.No. 10(b)

Simulate RIP using CISCO packet tracer.

Initial Ip configuration

Device	Interface	Ip Configuration	Connected with
PC0	Fast Ethernet	10.0.0.2/8	Router0's fa0/1
Router0	Fa0/1	10.0.0.1/8	PC0's Fast Ethernet
Router1	S0/0/1	192.168.1.254/30	Router2's S0/0/1
Router2	S0/0/0	192.168.1.253/30	Router1's S0/0/1
PC1	Fast Ethernet	20.0.0.2/30	Router2's Fa0/1

Assign Ip address to PC1.

Double click PC1 → Desktop → Ip configuration

Assign the Ip address.

Double click the Router0, click CLI

Press Enter.. to access command prompt of Router0

Following commands are used to access the global configuration

Router > enable

Router # configure

terminal

Enter terminal command, one per line End with

CNTL/Z.

Following commands assign Ip address as fast Ethernet 0/0

Router (config) # interface fastEthernet 0/0

Router (config-if) # ip address 10.0.0.1

255.0.0.0 Router (config-if) # no shutdown

Router (config-if) # exit

Router (config) #

interface fastEthernet 0/0 command is used to enter the interface mode

no shutdown - will bring interface up

exit - used to return in global configuration mode.

Router:

Router > enable

Route # configure

terminal

Enter configuration commands, one per line And with CNTL/Z.

Router (config) # interface serial 0/0/0

Router (config-if) # no shutdown

Router (config-if) # exit

Router (config) interface serial 0/0/1

Router (config-if) # ip address 192.168.1.246

255.255.255.252.

Router (config-if) clock rate 64000

Router (config-if) # bandwidth 64

Router (config-if) # no shutdown

exit

using these commands we can also assign ip address to router2.

Configure RIP Routing protocol.

Router 2 step.

(*) Enable RIP Routing protocol from global configuration mode.

(*) Tell RIP routing protocol from global Configuration mode / you want to advertise.

Router rip - tell router to enable

network - Specifies network which we want to advertise.

Router 0 :

Router0 (config) # router rip

Router0 (config-router) # network 10.0.0.0

Router 0 (config - Route) # network 192.168.1.252.

Router 1 :

Router 1 (config) # Router rip

Router 1 (config - routers) # network 192.168.1.244

Route (config) # network 192.168.1.248.

Router 2

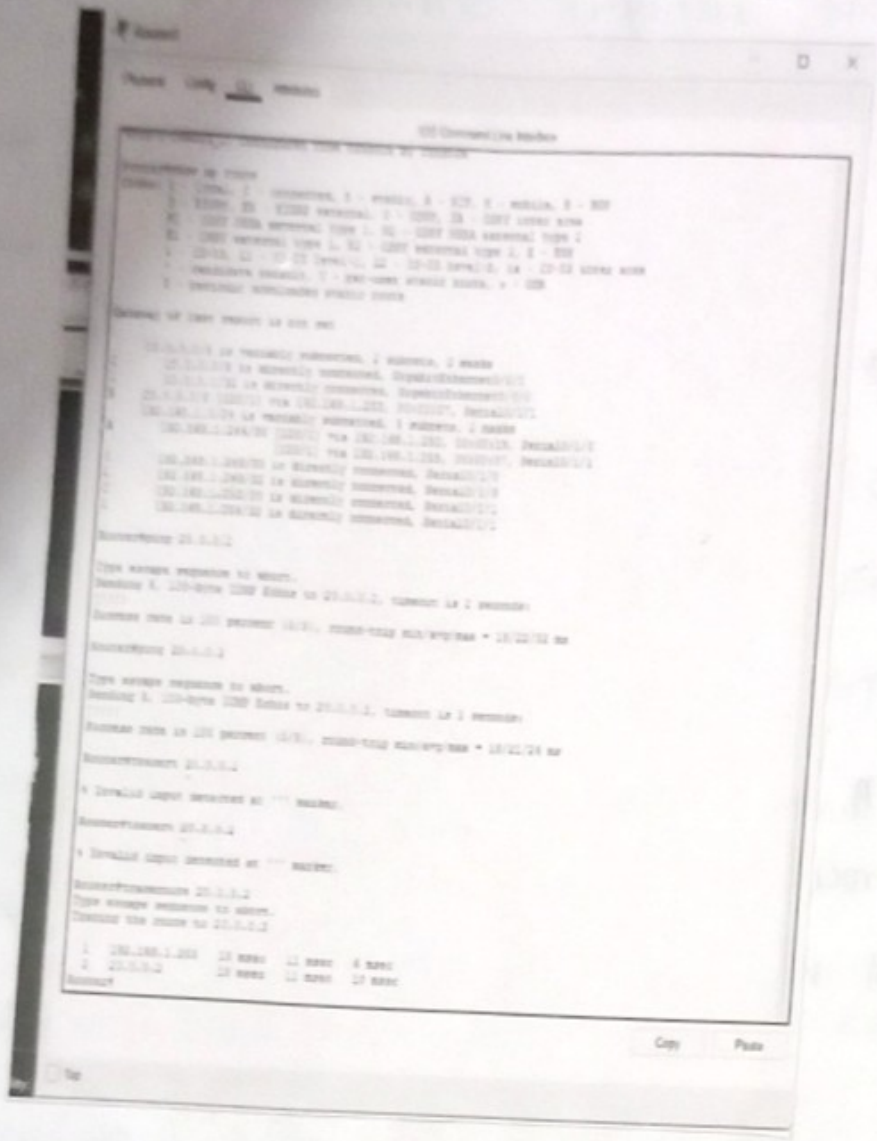
Router 2 (config) # router rip

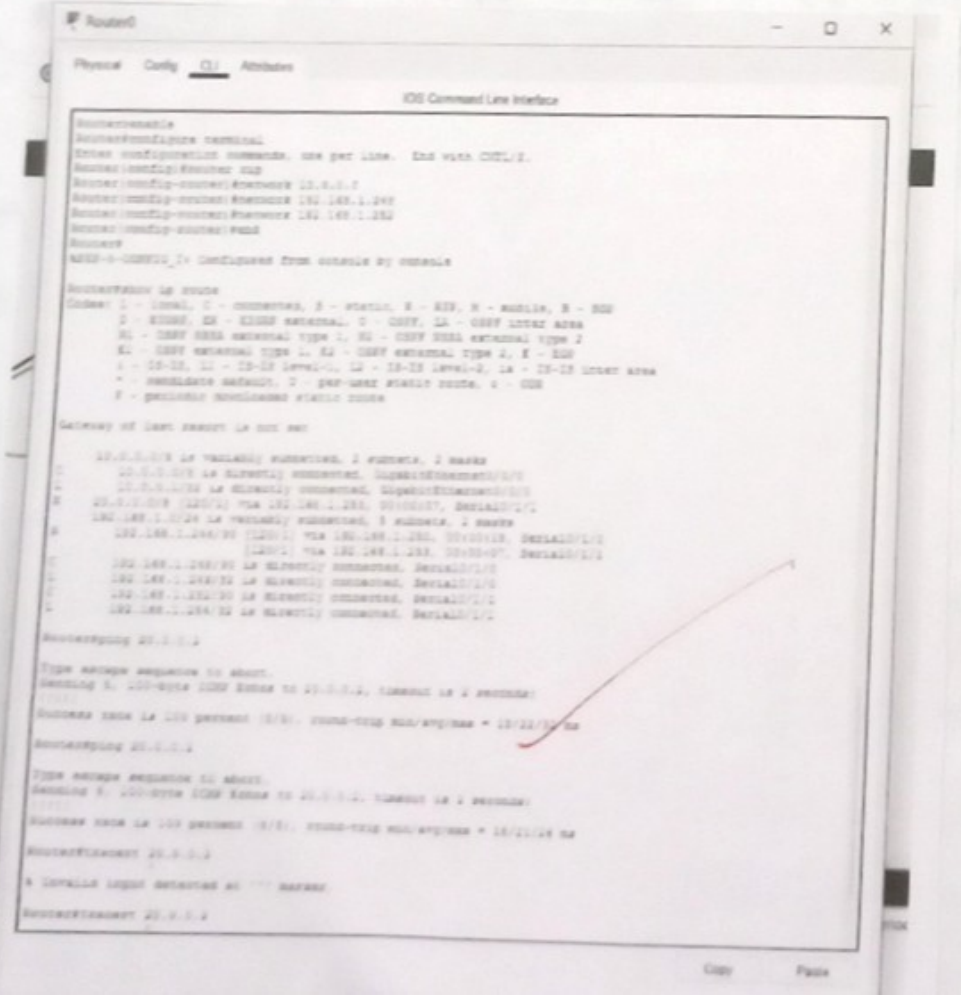
Router 2 (config - router) # network 20.0.0.0.

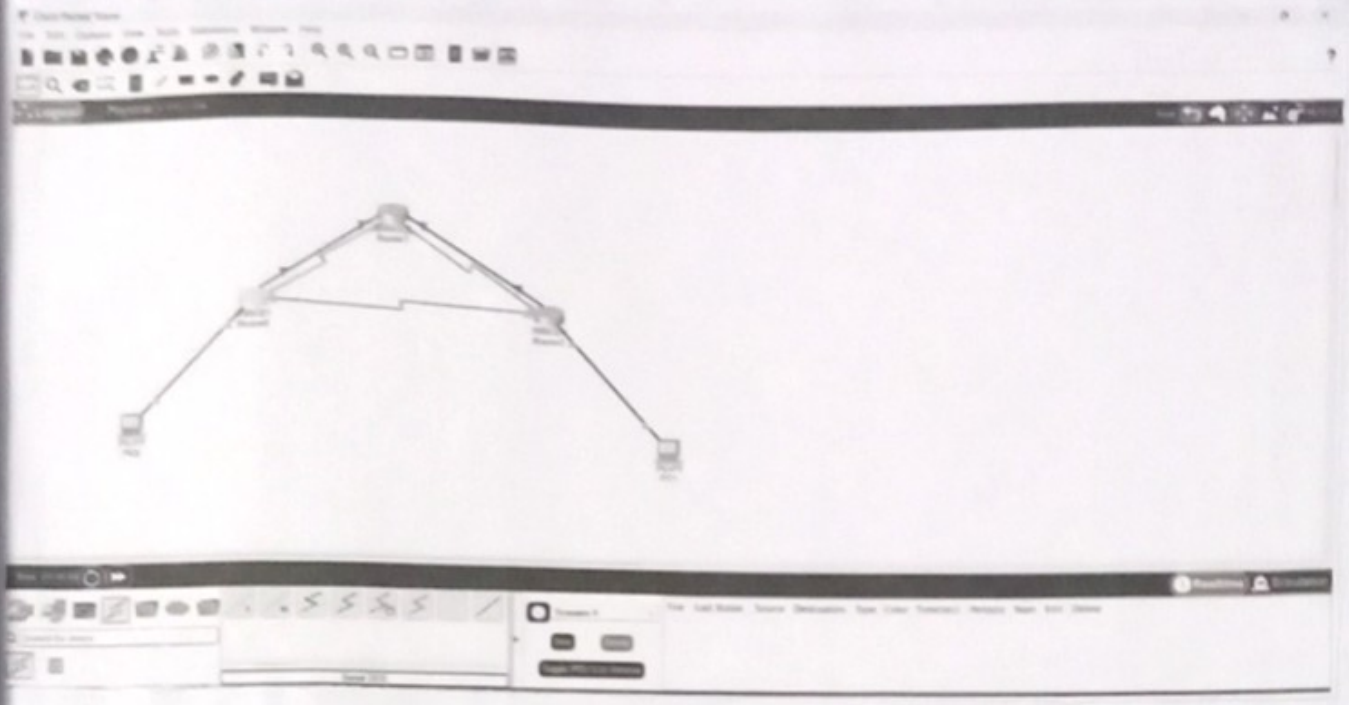
Router 2 (config - router) # network 192.168.1.252.

Router 2 (config - route) # network 192.168.1.244

Result:




Physical
Config
CLI
Attributes
IOS Command Line Interface
Copy
Paste



Result:

Therefore RIP has been executed successfully in Cisco packet tracer